



EnterpriseAI

Credit-Based Usage System

Product Requirements Document - Version 2.0

Updated for Decision Engine, Live Business Plan and Deterministic-First Simulation

Product	EnterpriseAI
Feature	AI Credit Management and Subscription Usage Control
Version	2.0
Priority	High
Prepared for	Development Team
Supersedes	Credit-Based Usage System PRD v1.0
Target plans	Explorer, Starter Insight, Decision Engine
Date	19 August 2026

Area	v1.0	v2.0 update
Plans	Explorer + Starter Insight	Adds Decision Engine with 2,000 monthly AI credits and 3-user entitlement.
Scenario simulation	10 credits per scenario	AI interpretation/recommendation is charged separately.
Live Business Plan	Not included	Adds import/extraction, narrative refresh, variance interpretation, decision summaries and monthly AI review.
Integrations	Not defined as credit rule	Existing connected-data synchronisation is explicitly 0 credits.
Credit logic	AI outputs consume credits	Clarifies that only new AI-generated content/interpretation consumes credits; deterministic intelligence remains free.
Plan configuration	Explorer/Starter allocations	Adds Decision Engine allocation, entitlements and future-proof plan configuration.
Analytics	AI cost and usage	Adds deterministic-vs-AI usage analytics, Live Plan AI spend and Decision Engine margin metrics.

Core v2 Principle

Credits represent consumption of AI generation/interpretation resources - not access to deterministic business intelligence. Calculations, ontology propagation, KPI updates, connected-data synchronisation, deterministic scenario execution and plan versioning must not consume AI credits.

1. Product Overview

EnterpriseAI currently controls AI-powered usage through a unified credit wallet. Version 2 extends the existing architecture to support the Decision Engine subscription tier and the Live Business Plan while preserving predictable AI infrastructure costs.

The system must distinguish between deterministic computation and AI generation. EnterpriseAI's Viability, Survival, Stability, Growth, Fragility/Master Score, ontology propagation, KPI calculations and deterministic scenario calculations are product intelligence functions and should not consume AI credits merely because they run.

Credits are consumed only where EnterpriseAI requests new AI-generated content, interpretation, narrative explanation or AI-assisted extraction.

2. Objectives

1. Give users flexibility to use AI credits across eligible AI-powered features.
2. Control and predict EnterpriseAI AI-provider costs.
3. Keep deterministic business intelligence accessible without artificial usage friction.
4. Support Explorer, Starter Insight and Decision Engine subscriptions.
5. Protect the Decision Engine gross margin while making its 2,000-credit allowance feel generous.
6. Support Live Business Plan AI operations without charging for ordinary KPI monitoring or connected-data sync.
7. Prevent excessive, abusive or duplicate AI usage.
8. Support monthly and annual subscriptions with monthly credit resets.
9. Allow future top-ups without redesigning the wallet.
10. Maintain a complete, auditable transaction history.
11. Deduct credits only after successful AI generation.
12. Allow administrators to change feature credit costs and plan allocations without code changes.

3. Subscription Plans

3.1 Explorer Plan

Price: £0

Billing type: Free

Credit allocation: 50 one-time AI credits

Reset: No monthly refresh

Users: 1

- 50 one-time AI credits
- Unlimited workspace and non-AI financial tools
- Unlimited products, customers and vendors
- Unlimited invoices and quotations
- Business Registration Guide
- 1 marketplace listing
- 1 user

Explorer credits are issued once per verified account and remain available while the account remains active, subject to abuse controls.

3.2 Starter Insight Plan

Monthly price: £19/month

Annual price: £190/year (marketed as 17% annual saving; rounded)

Credit allocation: 500 AI credits per billing month

Users: 1

- 500 AI credits every month
- Idea validation
- Business plan generation
- Business proposal generation
- Sales letter generation
- Basic scenario access
- Basic Fragility / engine interpretation where enabled
- Unlimited products, customers and vendors
- Unlimited invoices and quotations
- 1 marketplace listing
- 1 user

Annual subscribers receive 500 credits monthly. Do not issue the full annual allocation upfront.

3.3 Decision Engine Plan

Monthly price: £59/month

Annual price: £590/year (marketed as 17% annual saving; rounded)

Credit allocation: 2,000 AI credits per billing month

Users: 3

- Everything available in Starter Insight subject to plan entitlements
- 2,000 AI credits every month
- Live Business Plan
- Plan-vs-Actual KPI monitoring
- Variance detection and deterministic risk flags
- Advanced scenario simulation library
- Multi-scenario comparison
- Sensitivity analysis, breakpoint detection and safe-zone calculations where implemented
- Viability, Survival, Stability and Growth intelligence
- Fragility / Master Business Intelligence Score
- Adaptive scenario recommendations
- Scenario-to-Live-Plan decision adoption
- Decision outcome monitoring
- Use of already-built third-party integrations as Live Plan/ontology data sources
- 3 users
- Up to 5 marketplace listings (commercial configuration; admin-editable)



Decision Engine Credit Rule

The plan provides substantially more AI capacity, but the intelligence engine itself must not become pay-per-calculation. Deterministic simulations, KPIs, alerts, scores and connected-data sync remain 0 credits. AI explanation is metered separately.

3.4 Plan Allocation Summary

Plan	Monthly price	Annual price	AI credits	Reset	Users
Explorer	£0	£0	50 one-time	None	1
Starter Insight	£19	£190	500/month	Monthly	1
Decision Engine	£59	£590	2,000/month	Monthly	3

4. Credit Control Principles

- Only actions that create new AI-generated output or AI-assisted extraction should consume credits.
- Deterministic calculations must not consume credits.
- Connected-system synchronisation must not consume credits.
- Reading, editing, viewing, saving, exporting or downloading existing outputs must not consume credits.
- Credits must be held before an AI call and committed only after a valid result is successfully saved.
- Credit values must be backend-configurable and never permanently hardcoded in the UI.
- Plan eligibility and credit cost are separate checks: a user may have credits but still lack access to a feature.
- One AI request must map to one unique idempotency key.

5. AI Credit Costs by Feature

The following values are the initial v2 configuration. All values must remain editable through admin configuration.

Feature code	Feature name	Credits	Minimum plan	Credit controlled	Refund on failure
idea_validation	Idea Validation	5	Explorer	Yes	Yes
market_data_refresh	AI Market Data Refresh / Synthesis	4	Starter Insight	Yes	Yes
business_plan_full	Complete Business Plan	40	Explorer	Yes	Yes
business_plan_section	Regenerate Business Plan Section	3	Explorer	Yes	Yes
proposal_full	Complete Business Proposal	25	Starter Insight	Yes	Yes
proposal_section	Regenerate Proposal Section	2	Starter Insight	Yes	Yes
sales_letter_full	Complete Sales Letter	10	Starter Insight	Yes	Yes
sales_letter_section	Regenerate Sales Letter Section	1	Starter Insight	Yes	Yes
fragility_ai_interpretation	AI Fragility / Score Interpretation	3	Starter Insight	Yes	Yes
scenario_ai_interpretation	AI Scenario Interpretation	3	Starter Insight	Yes	Yes
scenario_ai_recommendation	AI Scenario Recommendation Explanation	3	Decision Engine	Yes	Yes
scenario_comparison_ai_summary	AI Multi-Scenario Comparison Summary	5	Decision Engine	Yes	Yes
sensitivity_ai_interpretation	AI Sensitivity / Breakpoint Interpretation	3	Decision Engine	Yes	Yes
live_plan_import_extract	AI Business Plan Import & Structured Extraction	10	Decision Engine	Yes	Yes
live_plan_section_refresh	Refresh One Live Plan Narrative Section	3	Decision Engine	Yes	Yes
live_plan_variance_interpretation	AI Variance Interpretation	3	Decision Engine	Yes	Yes
decision_ai_summary	AI Decision Summary	3	Decision Engine	Yes	Yes
decision_narrative_update	AI Narrative Update After Decision Adoption	5	Decision Engine	Yes	Yes



Feature code	Feature name	Credits	Minimum plan	Credit controlled	Refund on failure
live_plan_full_refresh	Full Live Business Plan Narrative Refresh	20	Decision Engine	Yes	Yes
live_plan_monthly_review	AI Monthly Business Review	10	Decision Engine	Yes	Yes
integration_ai_insight	AI Interpretation of Connected-Data Insight	3	Decision Engine	Yes	Yes

6. Deterministic Actions That Must Cost 0 Credits

- Run a deterministic scenario simulation using existing formulas and engine logic.
- Recalculate Viability, Survival, Stability or Growth scores.
- Recalculate Fragility / Master Business Intelligence Score.
- Run deterministic multi-scenario comparisons.
- Run deterministic sensitivity analysis.
- Run breakpoint detection.
- Run safe-zone calculations.
- Propagate scenario changes through the ontology calculation layer.
- Calculate or refresh Live Business Plan KPIs.
- Calculate Plan-vs-Actual variance.
- Generate deterministic risk flags.
- Generate rules-based alerts.
- Create or compare Live Plan versions.
- Adopt an approved scenario into the Live Business Plan structured plan state.
- Monitor an adopted decision against actual outcomes.
- Synchronise or normalise data from QuickBooks, Zoho, Xero, Stripe, PayPal or other existing connectors.
- Create/edit customers, vendors, products, invoices, quotations or expenses.
- Use standard calculators and deterministic scoring.
- View/edit/save/export/download an existing output.

Important

Do not charge 10 credits merely because the user clicks “Run Scenario”. In v2, the deterministic simulation itself is free. If the user requests an AI-written explanation of the simulation, charge the configured AI interpretation cost.

7. Live Business Plan Credit Rules

Live Plan action	Credits	Reason
Adopt existing structured EnterpriseAI Business Plan	0	No new AI generation if structured plan data already exists.
Import PDF/DOCX and use AI to extract structured assumptions	10	AI extraction operation.
Map confirmed assumptions to ontology / KPIs	0	Deterministic data operation.
Plan-vs-Actual KPI refresh	0	Deterministic.
Variance calculation and severity classification	0	Deterministic.
Rules-based alert	0	Deterministic.
AI explanation of a material variance	3	New AI-generated interpretation.
Adopt scenario into future planned state	0	Structured state/version operation.
Create new immutable Live Plan version	0	Database operation.
Refresh one affected narrative section	3	New AI text generation.
Update narrative after adopted decision	5	AI may need to update multiple linked sections.
Refresh all Live Plan narrative sections	20	Large AI generation operation.
Monthly AI Business Review	10	Consolidated AI narrative based on deterministic results.



8. Decision Engine Scenario Credit Rules

Scenario access is controlled by subscription entitlement. Credit charging is controlled independently by whether AI generation is requested.

Action	Credit cost	Implementation rule
Run deterministic single scenario	0	Execute rules/formulas only.
Run deterministic multi-scenario comparison	0	No AI narrative required.
Sensitivity sweep	0	Deterministic range calculation.
Breakpoint detection	0	Deterministic threshold calculation.
Safe-zone identification	0	Deterministic output.
AI scenario interpretation	3	Call AI only after deterministic results exist.
AI recommended-action explanation	3	Explain a rules-selected recommendation.
AI multi-scenario summary	5	Summarise multiple deterministic scenario outputs.
AI sensitivity/breakpoint explanation	3	Interpret deterministic sensitivity output.
Scenario adoption	0	No AI required.
Decision outcome monitoring	0	Compare expected vs actual.
AI decision performance commentary	3	Optional AI explanation if exposed in UI.

9. Third-Party Integration Credit Rules

EnterprateAI already supports connection to third-party systems. The credit system must treat these integrations as data sources, not AI features.

- Connecting or disconnecting a provider: 0 credits.
- OAuth/authentication: 0 credits.
- Background or manual data sync: 0 credits.
- Normalising connected data into the ontology: 0 credits.
- Recalculating KPIs or engine scores because connected data changed: 0 credits.
- Generating an AI interpretation specifically from connected-data intelligence: charge integration_ai_insight (initially 3 credits).

10. Credit Deduction Rules

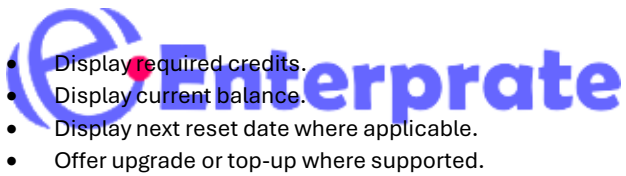
10.1 Successful AI Generation

1. User initiates AI-powered action.
2. Validate session, subscription and feature entitlement.
3. Retrieve configured credit cost.
4. Check available balance.
5. Create generation request and unique idempotency key.
6. Place required credits on hold.
7. Call AI provider.
8. Validate output.
9. Persist output.
10. Commit held credits.
11. Return output and updated balance.

Credits must not be permanently deducted when the provider errors, times out, returns an invalid/empty result, the application fails before saving, or a permission/payment error prevents completion.

10.2 Insufficient Credits

- Do not start the AI provider request.



Example

This AI explanation requires 5 credits. You currently have 2 credits. Your credits reset on 1 September 2026. Upgrade or purchase additional credits to continue.

10.3 Duplicate Submission Protection

- Double-clicks
- Browser refresh
- Network retries
- Duplicate API requests
- Repeated webhook delivery
- Repeated frontend mutation retries

Every AI generation request must have a unique idempotency key. The same key must never commit more than one deduction.

11. Credit Expiry and Reset Rules

11.1 Explorer

- 50 credits issued once after verification.
- No reset.
- No new allocation after cancellation/recreation if already issued.

11.2 Starter Insight - Monthly

- 500 credits issued at billing-cycle start.
- Unused subscription credits expire at next reset.
- No rollover by default.

11.3 Starter Insight - Annual

- 500 credits issued monthly based on subscription anniversary.
- Do not issue 6,000 upfront.
- Unused monthly credits expire at next reset.

11.4 Decision Engine - Monthly

- 2,000 credits issued at billing-cycle start.
- Unused subscription credits expire at next reset.
- No rollover by default.

11.5 Decision Engine - Annual

- 2,000 credits issued monthly based on annual subscription anniversary.
- Do not issue 24,000 upfront.
- Unused monthly credits expire at next reset.

11.6 Upgrades

Upgrade	Behaviour
Explorer → Starter	Issue 500 Starter credits immediately after successful payment. Existing promotional Explorer credits may remain for 30 days if configured.
Explorer → Decision Engine	Issue 2,000 Decision Engine credits immediately after successful payment.
Starter → Decision Engine	Activate Decision Engine immediately. Recommended default: set paid-plan allowance to 2,000 for the new cycle and preserve top-up credits. Do not double-allocate full Starter + Decision monthly pools.



Upgrade proration behaviour must be configurable and aligned to Stripe subscription handling. Subscription entitlements begin only after the payment state required by billing rules is satisfied.

11.7 Downgrades

- Maintain paid-plan access until end of paid period unless billing policy requires otherwise.
- Unused subscription credits expire at end of applicable paid period.
- Top-up credit treatment follows top-up rules.
- No new Explorer allocation if free allocation was already received.

12. User Interface Requirements

12.1 Credit Balance

- Dashboard header
- Account menu
- Subscription page
- AI-generation screens
- Live Business Plan AI actions
- Scenario AI interpretation screens
- Mobile navigation

Display example
1,742 of 2,000 AI credits remaining
Resets 1 September 2026

12.2 Pre-Generation Cost Disclosure

- Generate Business Plan - 40 credits
- Interpret Scenario with AI - 3 credits
- Refresh Live Plan Section - 3 credits
- Monthly AI Business Review - 10 credits

Zero-credit deterministic actions should not display confusing “0 credits” confirmations everywhere. A lightweight “No AI credits used” tooltip may be shown where useful, especially around scenario simulation.

12.3 Pricing Page

Explorer	Starter Insight	Decision Engine
£0	£19/month	£59/month
50 one-time AI credits	500 AI credits/month	2,000 AI credits/month
1 user	1 user	3 users
Entry-level AI generation	Planning + AI content	Live Plan + advanced decision intelligence
Operational tools	Operational tools	Operational tools + advanced plan/decision workflows

Annual pricing: Starter £190/year and Decision Engine £590/year, presented as approximately 17% annual saving. Pricing must be commercial configuration, not embedded in AI-credit logic.

12.4 Usage History

Date	Feature	Action	Credits	Status	Balance
19 Aug 2026	Scenario	Deterministic simulation	0	Completed	2,000
19 Aug 2026	Scenario	AI interpretation	-3	Completed	1,997

Date	Feature	Action	Credits	Status	Balance
19 Aug 2026	Live Plan	Section refresh	-3	Completed	1,994
20 Aug 2026	Integration	QuickBooks sync	0	Completed	1,994

13. Credit Transaction Types

- initial_allocation
- monthly_allocation
- subscription_renewal
- promotional_credit
- ai_generation_deduction
- credit_hold
- credit_hold_release
- failed_generation_refund
- manual_admin_adjustment
- top_up_purchase
- subscription_upgrade
- subscription_downgrade
- expiry
- chargeback_adjustment

Zero-credit deterministic activity should be captured in product analytics/event logs where needed, but it does not need to create a financial credit-ledger transaction unless the team wants a unified activity history. If shown in the user Usage History, it may be sourced from the activity log rather than the credit ledger.

14. Recommended Data Model

14.1 credit_wallets

Field	Type
id	UUID
user_id	UUID
workspace_id	UUID
plan_id	UUID
available_credits	Integer
held_credits	Integer
lifetime_credits_issued	Integer
lifetime_credits_used	Integer
next_reset_at	Timestamp
last_reset_at	Timestamp
created_at	Timestamp
updated_at	Timestamp

14.2 credit_transactions

Field	Type
id	UUID
wallet_id	UUID



Field	Type
user_id	UUID
workspace_id	UUID
feature_code	String
generation_id	UUID nullable
transaction_type	Enum
credits	Integer
balance_before	Integer
balance_after	Integer
status	Enum
idempotency_key	String unique
description	Text
expires_at	Timestamp nullable
created_at	Timestamp

14.3 credit_feature_config

Field	Type
id	UUID
feature_code	String unique
feature_name	String
credit_cost	Integer
enabled	Boolean
minimum_plan	String
credit_controlled	Boolean
refundable_on_failure	Boolean
created_at	Timestamp
updated_at	Timestamp

14.4 plan_credit_config

Field	Type
id	UUID
plan_code	String
allocation_type	Enum
credits_per_period	Integer
reset_frequency	Enum
rollover_enabled	Boolean
expiry_enabled	Boolean
enabled	Boolean



Field	Type
created_at	Timestamp
updated_at	Timestamp

14.5 Recommended plan_feature_entitlements

Add or reuse a plan-entitlement layer so access control is not inferred from credit balance.

Field	Type
id	UUID
plan_code	String
feature_code	String
enabled	Boolean
limit_value	Integer nullable
limit_period	Enum nullable
created_at	Timestamp
updated_at	Timestamp

Why this matters

A Decision Engine user may have 2,000 credits, but Starter-only or future Strategic Intelligence features should still be controlled through plan entitlements. Credits answer “can the user afford this AI action?” Entitlements answer “does this plan include the feature?”

15. API Requirements

15.1 Get Credit Balance

```
GET /api/credits/balance

{
  "plan": "decision_engine",
  "available_credits": 1742,
  "held_credits": 0,
  "monthly_allocation": 2000,
  "next_reset_at": "2026-09-01T00:00:00Z"
}
```

15.2 Get Feature Costs

```
GET /api/credits/features
```

15.3 Check Availability

```
POST /api/credits/check

{ "feature_code": "scenario_ai_interpretation" }

{
  "allowed": true,
  "entitled": true,
}
```



```
"credit_controlled": true,  
"required_credits": 3,  
"available_credits": 1742,  
"remaining_after_action": 1739  
}
```

15.4 Reserve / Commit / Release

- POST /api/credits/reserve
- POST /api/credits/commit
- POST /api/credits/release

15.5 Usage History

```
GET /api/credits/transactions  
Filters: start_date, end_date, feature_code, transaction_type, status, pagination
```

15.6 Deterministic Feature Check

For a feature with `credit_controlled=false`, the API should still enforce `minimum_plan/entitlement` but should return `required_credits=0` and must not create a credit hold.

16. AI Generation Workflow

```
User requests AI action  
↓  
Validate session + plan entitlement  
↓  
Load credit_feature_config  
↓  
If credit_controlled = false → execute feature with no wallet mutation  
↓  
If credit_controlled = true → check balance  
↓  
Create generation + idempotency key  
↓  
Reserve credits  
↓  
Call AI provider  
↓  
Validate response  
↓  
Save output  
↓  
Commit credits  
↓  
Return output + updated balance
```

17. Deterministic Scenario Workflow

```
User clicks Run Scenario  
↓  
Validate plan entitlement  
↓  
Load ontology/base state  
↓  
Apply ScenarioChange in temporary simulation state  
↓  
Run Viability / Survival / Stability / Growth / Master  
↓  
Persist SimulationResult  
↓  
Return before-vs-after metrics
```



↓
Credits used = 0

Optional: user clicks Explain with AI
↓
Reserve configured AI credits
↓
Generate interpretation
↓
Commit after success

18. Live Business Plan Credit Workflow

Live Plan structured update / KPI refresh / variance calculation
↓
No AI call
↓
Credits used = 0

If user requests AI extraction / narrative / interpretation
↓
Check feature entitlement
↓
Reserve configured credits
↓
Call AI
↓
Validate + save
↓
Commit credits

19. Top-Up Credits

The v1 top-up concept remains valid and should support all paid plans when enabled.

Package	Price
100 AI credits	£4.99
300 AI credits	£11.99

- Add only after confirmed payment.
- Do not expire at normal monthly reset unless commercial policy changes.
- Consume subscription credits before top-up credits where technically practical.
- Retain source in ledger even if displayed as a unified balance.
- Explorer users should upgrade before top-ups unless explicitly configured otherwise.

20. Stripe Billing Requirements

- checkout.session.completed
- invoice.paid
- invoice.payment_failed
- customer.subscription.created
- customer.subscription.updated
- customer.subscription.deleted
- charge.refunded
- charge.dispute.created

Billing event	Required credit action
Starter monthly payment succeeds	Activate Starter and allocate/reset to 500 subscription credits.



Billing event	Required credit action
Starter annual payment succeeds	Activate annual Starter; issue 500 credits and schedule monthly anniversary resets.
Decision monthly payment succeeds	Activate Decision Engine and allocate/reset to 2,000 subscription credits.
Decision annual payment succeeds	Activate annual Decision Engine; issue 2,000 credits and schedule monthly anniversary resets.
Payment fails	Do not issue new credits; apply grace rules and notify user.
Subscription cancelled	Preserve access to paid-period end where applicable; stop future allocations; downgrade at period end.
Upgrade Starter → Decision	Apply configured proration; activate Decision entitlement and correct allowance without duplicate full-cycle allocation.

All webhook processing must remain idempotent.

21. Admin Requirements

- View user credit balance and transaction history.
- Add/remove/refund credits with reason and audit trail.
- Edit AI credit cost by feature.
- Enable/disable credit-controlled features.
- Change minimum plan and credit_controlled flag.
- Edit plan allocations (50 / 500 / 2,000) without code changes.
- View high-usage accounts.
- View failed AI generations.
- View actual AI cost by feature and plan.
- View deterministic usage separately from AI usage.
- View Live Business Plan AI credit consumption.
- View Decision Engine AI gross margin.
- Export usage reports.
- Search by user email/workspace.

22. Analytics and Reporting

Metric	Description
Credit utilisation rate	$\text{Credits used} \div \text{allocated credits}$
AI cost per active user	$\text{AI provider cost} \div \text{active users}$
Revenue-to-AI-cost ratio	$\text{Subscription revenue} \div \text{AI cost}$
AI gross margin by plan	$\text{Plan revenue less attributable AI cost}$
Feature cost variance	Actual AI cost compared with configured credit cost
Credit exhaustion rate	Users reaching zero subscription credits
Top-up conversion rate	Eligible users buying top-ups
Deterministic scenario runs	Scenario runs that consumed 0 credits
Scenario AI interpretation rate	$\text{AI interpretations} \div \text{deterministic scenario runs}$
Live Plan AI usage	Credits used by Live Plan AI actions
Decision Engine credit utilisation	Average 2,000-credit pool used per active Decision subscriber
AI monthly review adoption	Users requesting monthly AI business review

23. Migration Requirements

23.1 Existing Explorer Users

- Preserve existing data/documents.
- Issue 50 one-time credits if not already issued.
- Do not reissue if previously allocated.

23.2 Existing Starter Users

- Preserve current 500-credit cycle.
- Migrate scenario_simulation configuration from 10 credits to deterministic 0-credit execution.
- Retain AI interpretation as a separate chargeable feature.
- Set next reset to existing billing date.

23.3 Decision Engine Users / Launch Migration

- Create Decision Engine plan configuration.
- Set 2,000 credits per monthly cycle.
- Create entitlements for Live Business Plan and advanced decision features.
- Existing generated documents remain accessible without charges.
- Existing connected integrations continue unchanged; do not re-authorise solely for Live Plan credit changes.

24. Security and Abuse Protection

- Email verification before Explorer allocation.
- Rate limiting on AI endpoints.
- Server-side plan entitlement checks.
- Server-side credit validation.
- Idempotent credit mutations.
- Fraud controls for repeated free-account creation.
- Atomic balance updates and database locking as appropriate.
- No negative balances.
- Role-based admin access.
- Complete transaction audit trail.
- Protect against manipulating the frontend to mark an AI action as deterministic.
- Feature credit_controlled/minimum_plan configuration must be trusted only from backend records.

25. Error Handling

Code	Meaning
INSUFFICIENT_CREDITS	Not enough AI credits.
FEATURE_DISABLED	Feature disabled.
PLAN_NOT_ELIGIBLE	Current plan lacks entitlement.
CREDIT_RESERVATION_FAILED	Credits could not be reserved.
GENERATION_FAILED	AI generation failed.
DUPLICATE_REQUEST	Idempotency key already processed.
SUBSCRIPTION_INACTIVE	Paid subscription inactive.
CREDIT_CONFIGURATION_MISSING	No credit config found.
ENTITLEMENT_CONFIGURATION_MISSING	No plan entitlement config found.
INVALID_CREDIT_CONTROL_MODE	Feature configuration contradicts supported execution mode.



26. Acceptance Criteria

1. Explorer receives exactly 50 one-time AI credits.
2. Starter receives 500 AI credits per monthly cycle.
3. Decision Engine receives 2,000 AI credits per monthly cycle.
4. Annual users receive monthly allocations rather than full-year credits upfront.
5. Feature costs come from configurable backend records.
6. Plan entitlement is checked independently from credit balance.
7. Deterministic scenario execution consumes 0 credits.
8. Viability, Survival, Stability, Growth and Master/Fragility recalculation consumes 0 credits.
9. Connected-data sync consumes 0 credits.
10. Live Plan KPI/variance processing consumes 0 credits.
11. AI scenario interpretation deducts the configured credit amount only after success.
12. Live Plan AI extraction/narrative features deduct configured credits only after success.
13. Failed AI generations release held credits.
14. Duplicate requests never duplicate deductions.
15. Users can view balance and usage history.
16. Admin can change credit costs, allocations, minimum plan and credit_controlled flags without code changes.
17. Existing documents remain accessible without credit charges.
18. Monthly subscription credits reset correctly.
19. Stripe events allocate the correct credits for Starter and Decision Engine.
20. Database prevents negative credit balances.
21. Usage analytics are available by user, plan and feature.
22. Pricing page reflects Explorer, Starter Insight and Decision Engine.
23. Mobile and desktop show the same accurate balance.
24. Zero-credit deterministic actions do not accidentally create wallet deductions.
25. All credit transactions are auditable.

27. Out of Scope for v2 Initial Release

- Credit gifting between users
- Credit transfers between workspaces
- Team-shared credit pools beyond normal plan wallet behaviour
- Enterprise custom credit contracts
- Automatic credit rollover
- Referral credit rewards
- Country-specific credit pricing
- Real-time variable credit pricing based directly on token count
- Charging credits for deterministic calculations
- Provider-specific credits by integration source

28. Recommended Delivery Phases

Phase 1 - v2 Core Configuration

- Add Decision Engine plan configuration
- Add 2,000-credit monthly allocation
- Split deterministic scenario execution from AI interpretation
- Add Live Plan AI feature codes
- Add/confirm plan entitlements
- Update pricing/balance UI

Phase 2 - Live Plan and Decision Integration

- Wire Live Plan AI actions to credit service
- Wire scenario AI explanation to credit service
- Ensure ontology/KPI/simulation operations bypass wallet mutation
- Add zero-credit activity visibility where required

Phase 3 - Commercial Optimisation

- Top-ups
- Usage alerts
- Credit exhaustion campaigns
- AI cost and gross-margin dashboard
- Credit-cost tuning based on real provider cost
- Optional promotional credits

29. Developer-Ready credit_feature_config Seed

Seed these records through migration/seed scripts. Do not rely on frontend constants.

feature_code	feature_name	cost	credit_controlled	minimum_plan	refund_on_failure
idea_validation	Idea Validation	5	true	Explorer	true
market_data_refresh	AI Market Data Refresh / Synthesis	4	true	Starter Insight	true
business_plan_full	Complete Business Plan	40	true	Explorer	true
business_plan_section	Regenerate Business Plan Section	3	true	Explorer	true
proposal_full	Complete Business Proposal	25	true	Starter Insight	true
proposal_section	Regenerate Proposal Section	2	true	Starter Insight	true
sales_letter_full	Complete Sales Letter	10	true	Starter Insight	true
sales_letter_section	Regenerate Sales Letter Section	1	true	Starter Insight	true
fragility_ai_interpretation	AI Fragility / Score Interpretation	3	true	Starter Insight	true
scenario_ai_interpretation	AI Scenario Interpretation	3	true	Starter Insight	true
scenario_ai_recommendation	AI Scenario Recommendation Explanation	3	true	Decision Engine	true
scenario_comparison_ai_summary	AI Multi-Scenario Comparison Summary	5	true	Decision Engine	true
sensitivity_ai_interpretation	AI Sensitivity / Breakpoint Interpretation	3	true	Decision Engine	true
live_plan_import_extract	AI Business Plan Import & Structured Extraction	10	true	Decision Engine	true
live_plan_section_refresh	Refresh One Live Plan Narrative Section	3	true	Decision Engine	true
live_plan_variance_interpretation	AI Variance Interpretation	3	true	Decision Engine	true
decision_ai_summary	AI Decision Summary	3	true	Decision Engine	true
decision_narrative_update	AI Narrative Update After Decision Adoption	5	true	Decision Engine	true
live_plan_full_refresh	Full Live Business Plan Narrative Refresh	20	true	Decision Engine	true
live_plan_monthly_review	AI Monthly Business Review	10	true	Decision Engine	true
integration_ai_insight	AI Interpretation of Connected-Data Insight	3	true	Decision Engine	true
scenario_deterministic	Deterministic Scenario Simulation	0	false	Starter Insight	false
scenario_multi_compare_deterministic	Deterministic Multi-Scenario Comparison	0	false	Decision Engine	false
sensitivity_deterministic	Deterministic Sensitivity Analysis	0	false	Decision Engine	false
breakpoint_deterministic	Deterministic Breakpoint Detection	0	false	Decision Engine	false
safe_zone_deterministic	Deterministic Safe-Zone Identification	0	false	Decision Engine	false
live_plan_kpi_refresh	Live Plan KPI Refresh	0	false	Decision Engine	false
live_plan_variance	Live Plan Variance Calculation	0	false	Decision Engine	false
live_plan_scenario_adopt	Adopt Scenario into Live Plan	0	false	Decision Engine	false
integration_sync	Connected Data Synchronisation	0	false	Decision Engine	false
engine_recalculation	Viability/Survival/Stability/Growth Recalculation	0	false	Starter Insight	false

30. Developer-Ready plan_credit_config Seed

plan_code	allocation_type	credits_per_period	reset_frequency	rollover	expiry
explorer	one_time	50	none	false	false
starter_insight_monthly	monthly	500	monthly	false	true
starter_insight_annual	monthly	500	monthly_anniversary	false	true
decision_engine_monthly	monthly	2000	monthly	false	true

plan_code	allocation_type	credits_per_period	reset_frequency	rollover	expiry
decision_engine_annual	monthly	2000	monthly_anniversary	false	true

31. Developer Implementation Notes

- Existing scenario_simulation=10 configuration must be migrated/deprecated, not left active alongside scenario_deterministic=0, otherwise users may be double-charged.
- If the UI combines “Run + Explain” into one button, backend should execute deterministic simulation first, then reserve credits only for the AI explanation step.
- If a user declines AI interpretation, the deterministic result must remain fully usable.
- AI credit charging must never be tied to the source of business data. QuickBooks/Zoho/Xero/Stripe/PayPal data is simply ontology input.
- Feature code naming should remain stable once released because transaction history and analytics depend on it.
- Admin changes to credit cost must affect future requests only and must not rewrite historical transactions.
- Always store actual provider cost separately from user-facing credits so commercial credit pricing can be adjusted independently of token/provider pricing.

32. Final Product Requirement

Required Behaviour

EnterprateAI credits must meter AI generation, not business intelligence itself. Users on eligible plans must be able to calculate, monitor, simulate and propagate deterministic business effects without consuming AI credits. Credits are charged only when EnterprateAI creates new AI-generated content, interpretation, synthesis or AI-assisted extraction.

The platform must retain full administrative control of plan allocations, credit pricing, entitlements, feature availability and refund behaviour without routine code changes. Every AI credit transaction must remain transparent, auditable, idempotent and financially measurable.